

Team Number: _____ School & Team Name: _____

Student Names: _____

ROCKET A ROCKET B ← CHECK-IN (UP TO 2 ROCKETS)

Construction Parameters:

T	F	T	F	5.a.	Team properly wears class B eye protection at all times.
T	F	T	F	2.b.	Parachutes are attached to ping pong balls with tape only.
T	F	T	F	3.a.	Rocket pressure vessels are made from a single 1-liter or less plastic carbonated beverage bottle with a nozzle opening internal diameter of approx. 2.2 cm and a standard neck height from flange to bottle's opening of under 1.6 cm. The bottle label is presented.
T	F	T	F	3.b.	The structural integrity of the pressure vessel is not altered. This includes, but is not limited to: physical, thermal, or chemical damage (e.g., cutting, sanding, using hot or super glues, spray painting.)
T	F	T	F	3.c.	The nose of the rocket is rounded or blunt at the tip and designed such that when a standard bottle cap (~3.1 cm diameter x 1.25 cm tall) is placed on top of the nose, no portion of the nose touches the inside top of the bottle cap.
T	F	T	F	3.d.	Tape is the only thing used to attach fins and other components to the pressure vessel. No glues of any type are used on the pressure vessel. There is no metal anywhere on the rocket or parachute payload system.
T	F	T	F	3.e.	Fins and other parts added to the bottle are 5 cm or higher above the level of the bottle's opening.
T	F	T	F	3.f.	All energy imparted to the rocket originates from air pressure provided by the air pump through the launcher. Water, gases other than air, explosives, liquids, chemical reactions, pyrotechnics, electrical devices, elastic powered flight assists, throwing devices, remote controls, and tethers are not used.
T	F	T	F	3.g.	Mylar/ultrafilm materials are not used in any part of the parachute or rocket.
T	F	T	F	3.h.	Participants are able to answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org .
1. T	F	2. T	F	ALL CONSTRUCTION PARAMETERS ABOVE ARE MET (If F, DO NOT RUN and team gets participation points only.)	

PRACTICE LOG

3. Is the Practice Log complete, incomplete, or not present?

Complete

Incomplete (0.85x multiplier)

Not present (0.7x multiplier)

4.a. *Practice Log includes 5 or more parameters (pressure (psi); estimated peak flight height (feet); time aloft; and 2 more chosen by the team) with data for 10 or more test flights prior to the competition.*

(If Incomplete or Not Present, do not let team pressurize beyond ES determined safe pressure that ensures rocket doesn't reach ceiling.)

1ST LAUNCH

4. A B

5. ____:____.____

6. T F

7. T F

12. T F

2ND LAUNCH

8. A B

9. ____:____.____

10. T F

11. T F

← LAUNCH PERIOD (IN 5 MIN.)

Which rocket from above was used?

← **Median Time Aloft**
(minutes:seconds)

6.e.i. Parachute Payload System successfully separates from the Rocket. (If F, 0.9x multiplier)

6.e.iv. No part of the Rocket or Parachute Payload System hits the ceiling or any parts connected to the ceiling during the ascending portion of the launch. (If F, 0.25x multiplier)

General Rule: The team is disqualified. (Notify the team and their coach as soon as possible.)